Technical report — Swiss Heritage Factuality: does Apertus recognise Switzerland's landmarks?

Final labels are the frozen LLM judge's (`judge-v1`). Photos were screened by AI and the 20 % audit sample was graded by a second AI system; no human audit of the grades was done (see Annotation and Limitations).

We built a four-language (FR/DE/IT/RM) benchmark of 3,412 questions about 251 Swiss heritage landmarks from the federal inventory (KGS), stratified by language region and fame, with photo, second-view, text-only and trap variants and per-item licences. Apertus 1.5 70B almost never recognises a landmark from its photo, and when it commits to a name it is almost always wrong (360 of 364 committed names). Whether it commits at all depends on the question's language: on the same photos it mostly declines in French (11 % of name answers confidently wrong) but answers in Romansh (68 % of names and 97 % of places wrong), where it also relocates half of all non-Graubünden landmarks to Graubünden. Shown a photo too generic to identify, it declines every time.

Track: Apertus Readiness - Track 1B (Swiss Voices), dimension: core-task intelligence (factuality)
Event: Online
Team: Red Pandas — Christopher Yazigi, Giovanni Yazbeck, Alfredo Pio Yazbeck
Demo / data: dataset `huggingface.co/datasets/chrizzto/swiss-heritage-factuality` · code `github.com/ChristoYusj/aper swiss-landmarks`

Use of Apertus

- **Model:** `swiss-ai/Apertus-v1.5-70B` (main), `swiss-ai/Apertus-v1.5-8B` (secondary comparison)
- **How it is used:** evaluation (multimodal: photo + question)
- **Where it runs:** hosted CSCS inference endpoint (OpenAI-compatible); the harness accepts any OpenAI-compatible endpoint via `LLM_NAME`, `LLM_BASE_URL`, `LLM_API_KEY`

Use Case Description

A visitor, a teacher or a tourism chatbot points a phone at a Swiss chapel, castle or fountain and asks "what is this, where is it, when was it built?". This is a natural, high-volume use of a Swiss multimodal assistant, and it is exactly where a confident wrong answer is costly: it is believable, it is about a real place, and the person asking usually cannot check it. Switzerland's heritage is also a clean test of *localised* knowledge: the federal inventory of cultural property (KGS) lists thousands of objects, most of them known only locally, spread over four language regions of very unequal size.

We therefore measure **factuality and hallucination** (SimpleQA-style: correct / incorrect / not attempted) of Apertus 1.5 on Swiss heritage landmarks, and ask four questions fixed before the run (pre-registration commit `a73986e`, `docs/analysis_plan.md`): does Apertus invent more about obscure landmarks (H1); are wrong answers pulled toward the German-speaking majority (H2) or toward famous look-alikes (H2b); is the failure one of *recognition* (photo) rather than *knowledge* (name given) (H3); and does it accept a false premise in the question (H4)?

Dataset Summary

Landmarks	251 KGS heritage objects (A: 104, B: 139, local: 8) in 4 BFS language regions × 3 fame tiers (264 drawn; 13 removed by the photo screening)
Fame tiers	Wikipedia articles: low 0, mid 1–2, high ≥ 3 (21–24 per cell; Romansh region: smaller pools, 12–18, the high-fame cell a census)
Types	churches 108, castles 61, museums 30, civic 18, bridges 14, fountains 10, towers/gates 8, monuments 2
Photos	251 main photos (Wikidata P18), 161 second views, 17 low-information photos; 500 px, EXIF stripped, per-image licence

Questions	3,412: photo_main 2,276 (each photo in 4 languages), photo_alt 732, text_only 322, trap_premise 48, trap_lowinfo 34
Question types	Q1 name, Q2 municipality + canton, Q3 year built (67 landmarks with a precise, unconflicted year)
Languages	French, German, Italian, Romansh (Rumantsch Grischun); identical templates, explicit "if unsure, say so"
Answer keys	Wikidata (CC0) + BFS municipality register: name aliases (flagged if only valid with the town), current/former municipality names, canton names in 5 languages, year windows
Licences	questions/keys/responses CDLA-Permissive-2.0; photos keep their CC licence (TASL attribution per item)

Arms. photo_main: the photo, asked in all four languages (same image → language is the only change). text_only: the landmark's name is given and the place or year is asked (FR, DE), separating *recognition* from *knowledge*. trap_premise: the question asserts the town of a real look-alike in another municipality ("This building is in Rothrist. Which building is it?"). photo_alt (FR, DE): a second view of the same object, a memorisation hint. trap_lowinfo (FR, DE): a photo too generic to identify; declining is right.

Selection funnel (seed 20261013): 13,417 heritage objects with a photo → 4,153 eligible (public landmark types, a current municipality, a specific label, one language region) → stratified sample of 264 + 83 same-cell reserves for replacements.

Data Collection Method

Sources. Wikidata (QLever mirror, cached queries in the repository) for objects, names, inception dates and administrative location; the Federal Statistical Office municipality register (01-10-2026) for municipality numbers, cantons and language regions; Wikimedia Commons for photos and their licences.

Answer-key construction and cleaning (all rules in code, all decisions logged in CHANGELOG.md):

- Name aliases from Wikidata labels/aliases/sitelinks in EN/FR/DE/IT/RM/GSW. Aliases that are only a town name or only type words are dropped. A name is *non-distinctive* if, after removing type words, saints' names and the town, nothing is left ("Kapelle St. Antonius", "Kunsthaus Zürich"): it then counts only together with the right town. All 786 tokens that occur in the names were classified one by one (AI-assisted).
- Place aliases: the names of the current municipality in all languages (a district, region or former municipality is not the answer, 16 such Wikidata labels removed and logged); town names shared by two municipalities (e.g. Gossau) require the canton.
- Year: only preferred, unconflicted inception values with year precision (window ±5) or decade precision; museum founding dates are not used as build years.
- An independent script re-checked 25 random landmarks against live Wikidata (25/25).

Question validity rules (decided from the question text only): a text-only "where is it?" question is kept only if the name shown is distinctive in both FR and DE and does not contain the town; a text-only year question is dropped if the shown name contains a year. 170 items were removed this way after the run; their responses are kept in metadata/excluded_items.jsonl with the reason.

Annotation (docs/annotation_guidelines.md; deviations from the plan are logged in CHANGELOG.md).

- *Photo screening, done by AI (deviation: the plan had the team screen).* Rules: exclude recognisable people, readable plates, private interiors, personal text (never blur); drop a main photo that does not show the object; pick a second view of the same object; mark photos too generic to identify. Claude Opus 5.5 screened all 917 candidate photos; three further Claude instances re-checked, without the first reasons, the 491 photos the screening proposed to ship (privacy flags disagreed on 2); a second AI agent (Hermes) reviewed them again under the team lead's policy "if unsure, exclude"; the lead decided 6 edge-case photos on 4 landmarks (one revised later). 429 photos were shipped. **No person reviewed every shipped photo.** 13 landmarks were removed (privacy 4, wrong object 4, unclear 5); their 120 answers per model are kept in metadata/excluded_items.jsonl. No model answer was shown at any step.

- *Grading audit:* a seeded, stratified 20 % sample (546) of the 2,726 70B answers that existed before the screening (522 still in the dataset; the later `photo_alt/trap_lowinfo` items are not covered), with a blind labelling page (no grades shown; English machine gloss for non-French answers; 40 hidden repeats for self-consistency). No human labels were collected: the sample was graded by the second AI agent (526 scored, 20 left unscored as uncertain), reported as AI-vs-judge agreement, **not** as a human audit.

Consent and personal data. No person is a data subject: photos with recognisable people are excluded, EXIF is stripped. Annotators sign the consent form in `docs/consent_form_template.md` (pseudonymous IDs).

Dataset Details

- **Quality metrics:** answer-key audit (above), selection funnel, per-cell counts, item-validity rules, stability of answers at temperature 0 (below), judge–code agreement and an independent re-read of every disagreement (AI-assisted), and an AI grading of the 20 % audit sample (no human audit).
- **Diversity / edge cases:** four languages incl. Romansh; fame tiers; same-photo language contrast; text vs photo; false-premise traps with real look-alikes; low-information photos; ambiguous town names; non-distinctive names.
- **Contamination:** the Apertus pre-training data list contains no Wikimedia/landmark image set by name, but includes web-scale image–text corpora that may contain some of these photos; URL-level checks are infeasible for us (limitation). The `photo_main` vs `photo_alt` contrast is our indirect check: no advantage for the main photos (Results 4).
- **Ethics and licensing:** photos with NC/ND/GFDL-only/unknown licences are excluded; every image row carries author, licence, source URL and the modification made. Takedown via the dataset's discussion page.

Evaluation

Method

- **Labels:** SimpleQA three-way. *Hallucination rate* = incorrect / attempted.
- **Graders:** (1) a deterministic code grader on 100 % (alias matching with the distinctiveness and ambiguity rules, abstention and hedge detection in all four languages, a gazetteer of all Swiss cantons and current and former municipalities); (2) an independent open-weights judge, `openai/gpt-oss-120b` (not Apertus), on 100 %, pinned to one provider, temperature 0; prompt tuned on a 40-item dev set (labels proposed by Claude; the lead ruled on the 4 hard cases and accepted the other 36 without item-by-item review) (final prompt: 40/40 agreement, two runs identical) and frozen before the full run and the audit sample; final label = judge's.
- **Statistics:** cluster bootstrap over landmarks (2,000 resamples) for CIs; exact McNemar for paired contrasts; Holm correction across the registered tests; Wilson intervals for rates.

Evaluation setup and inference parameters

Temperature 0, top_p 1, max_tokens 300, seed 1234, one image (base64 JPEG) + one question, no system prompt. Every response is stored with a content hash of its request; `verify` replays the stored answers. **Stability:** 100 items re-asked once and a random 20 of them twice more: 75 % identical text, **99 % identical code label** ($\kappa = 0.98$; the one change is a vague hedge that, asked again, named two candidate cantons), and all 20 probe items kept their label over four answers: the endpoint is not bit-for-bit deterministic, but the grades are stable.

Results

1. Recognition from a photo almost never succeeds, and committed answers are almost always wrong.

photo_main (n per cell = 251)	FR	IT	DE	RM
Name: not attempted (declines, or too vague to check)	89 %	76 %	58 %	**32 %**
Name: confidently wrong	11 %	24 %	41 %	**68 %**
Place: confidently wrong	2 %	2 %	30 %	**97 %**

Overall, name correct 0.4 % (4 of 1,004), place correct 0.6 % (6 of 1,004), years almost never answered (2 of 268, both wrong). Among committed name answers 98.9 % are wrong (360 of 364). The same instruction ("if you are not sure, say so") produces honest refusals in French and confident fabrication in Romansh, on identical images. Many Romansh name answers are only a building type ("Chasa da baselgia", a church building): not attempted when the type fits the landmark, incorrect when it does not (SimpleQA: not attempted means "without contradicting the gold").

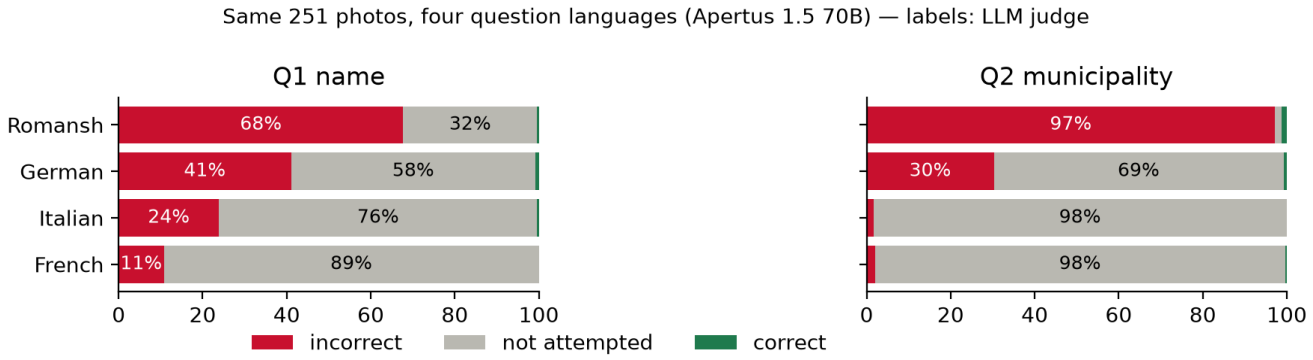


Figure 1: Same 251 photos, four question languages (Apertus 1.5 70B, LLM judge).

2. In Romansh the model moves landmarks to Graubünden (exploratory, location extracted with the gazetteer). Of 190 landmarks outside Graubünden, the Romansh answer places **96** in Graubünden; the French and Italian answers for the same photos place none there and the German answers 14 (paired McNemar RM vs FR/IT/DE, $p < 10^{-6}$). Overall 60 % of Romansh place answers say Graubünden (true share 24 %). The municipalities named are mostly German- and Italian-speaking Graubünden towns (93 of 137 named; e.g. Chur, Roveredo, Poschiavo): the pull is toward the canton associated with Romansh rather than toward Romansh-speaking villages. The 8B model shows the same pull, weaker: 35 % of its Romansh place answers say Graubünden, and it moves 54 of the 190 non-Graubünden landmarks there (70B: 96).

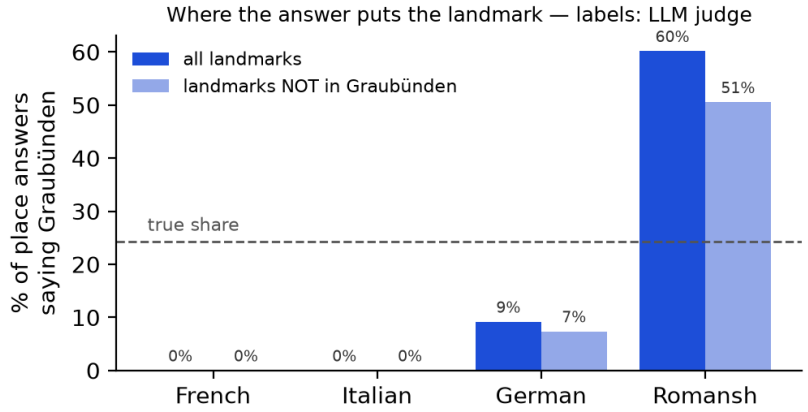


Figure 2: Share of place answers that say Graubünden, by question language (dashed: true share).

3. Famous names absorb unknown objects. Eleven different landmarks, mostly castles, are called "Castel Tarasp" (the iconic Engadine castle), eight Swiss landmarks become "Castle Vaduz" (Liechtenstein), and castles in French and Italian answers become "Château de Chillon". Disentis Abbey becomes "Kloster Einsiedeln".

4. A second view and an uninformative photo (descriptive, FR + DE). On the 161 landmarks with a second view, 70B names 2 landmarks correctly from the main photo and 1 other (in both languages) from the second view, and is wrong about as often (names 24 % on the second view vs 26 % on the main photo; places 11 % vs 16 %, McNemar $p = 0.04$). If the Wikidata main photos had been memorised, the model would do better on them than on other views; it does not (8B is even slightly better on the second view). On 17 photos too generic to identify, 70B declined all 34 times and 8B 33 of 34, although on the main photos of the same landmarks they gave wrong names 7 and 26 times: the models can tell an uninformative image from a recognisable one, but a photo that *looks* identifiable triggers a confident guess.

5. Registered hypotheses (Holm-adjusted):

	Result
H1 fame cliff (name)	not supported: hallucination 97.5 % (low) vs 99.2 % (high); floor effect
H1b fame cliff (place)	not supported (Δ = +2.3 pp, CI −1.0 to +6.4)
H2 majority-region pull	not supported: wrong towns for non-DE landmarks are 59 % German-speaking vs 65 % of all municipalities (registered comparison points the other way, p_Holm = 0.10)
H2b famous look-alike	not testable: 3 answers (1 landmark) name a look-alike's town
H3 seeing vs knowing (place)	**supported in French** (text 8 vs photo 0 correct of 95 landmarks, p_Holm = 0.04), same direction in German (9 vs 1, p_Holm = 0.09). Fragile: before the photo screening removed 13 landmarks, French was 8 vs 1 (p_Holm = 0.13); the one photo-correct French answer was on a landmark dropped for privacy
H4 false premise	**supported**: 79 % adopt the false town vs 29 % incorrect on the same photo asked plainly (p_Holm < 0.0001); counting a wrongly named building as incorrect too (team rule), 88 %

Knowing the name helps, and fame matters there: text-only place accuracy is 12.5 % for high-fame vs 3.7 % for low-fame landmarks, while from the photo both are ≈ 0 %. The bottleneck is visual recognition.

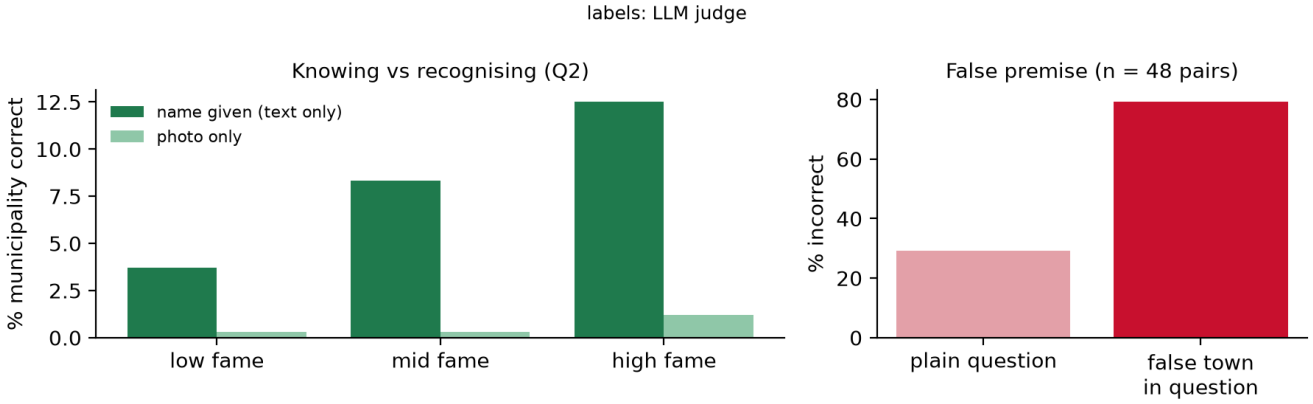


Figure 3: Left: municipality correct with the name given vs from the photo, by fame (H3). Right: incorrect on the plain photo question vs with a false town in the question (H4).

6. Smaller model. Apertus 1.5 8B declines less (name: 28 % not attempted vs 64 % for 70B) and is confidently wrong more often in every language (name wrong: FR 78 %, IT 48 %, DE 70 %, RM 90 %). Its fame effect on place questions points the expected way (H1b: 98.9 % vs 93.2 % hallucination, low vs high fame; bootstrap CI +1.3 to +10.4 pp) but is not significant after Holm (p_Holm = 0.07). Unlike 70B, 8B is not more misled by the false premise than by the plain question (71 % adopt the false town vs 83 % wrong on the control, n.s.): it is already wrong on the plain question. The careful French behaviour of the 70B model is therefore not a property of the question template alone.

7. Agreement and judge errors. Judge vs code grader: κ = 0.95 (70B; 97.9 % identical labels) and 0.94 (8B; 96.8 %). Every disagreement (180 now; 236 re-reads over two rounds, 224 still in the dataset; `disagreement_adjudication.jsonl`) was re-read independently against the written rules, with a second blind read where the judge looked wrong (AI-assisted, not a human audit): of the 224, the judge was right in 86, the code grader in 131, neither in 1, and 6 stayed unsettled. The judge's main error is strictness on vague answers whose building type fits the landmark ("church building" for a church), plus saint names without a town; it also leaves vague centuries outside the year window as not attempted. Replacing the judge's label wherever the re-read settled it lowers the Romansh name-wrong rate from 68 % to 59 % (70B; 8B 90 % → 81 %), raises the

small Romansh year cell by 6 pp and lowers the premise-trap rate from 88 % to 83 %; no other 70B language × question cell moves by more than about 1.5 pp, so the language pattern is not a grading artefact. The registered results keep the frozen judge's labels. *AI cross-check of the audit sample:* a second AI agent graded the 546 sampled answers (504 scored and still in the dataset): κ = 0.96 with the judge (98.2 % identical) and 0.99 with the code grader; lowest in Romansh (κ = 0.84). *Human audit:* not done, so the track's 20 % human check is not met (see Limitations).

Dataset Limitations

- Results are for one prompt template per language; Romansh templates were not reviewed by a native speaker (exploratory). The language effect could partly reflect template wording; a no-abstention-cue variant is future work.
- Gold comes from Wikidata; errors there propagate (mitigated by the rule-based checks, the 25-landmark live re-check and the AI-assisted re-reading of every grader disagreement against the gold). The photo screening also found one Wikidata main photo of the wrong building (removed).
- Fame is proxied by Wikipedia article counts. Low accuracy everywhere limits the power of fame contrasts (floor).
- Neither automatic grader is perfect. The frozen judge is too strict on vague answers that fit the landmark (see Agreement). The code grader (second rater) misses fine-grained contradictions: Romansh "house of X" phrases, a denomination the gold does not state, name-shaped phrases built from common nouns ("Tour de l'Horloge" for another tower), and descriptive museum names.
- Post-run corrections to the code grader and item validity were made after seeing answers; each is documented with its effect in `CHANGELOG.md` and pinned by regression tests. One independent review read every answer the code grader marked incorrect and moved 109 (70B) and 356 (8B) labels, mostly refusals phrased in ways the patterns did not know; a third read 831 labels and its fixes moved 14 (70B) and 29 (8B) more (every changed label was read individually; these reviews were AI-assisted, not a human audit).
- No human audit of the grades: the 20 % audit sample was graded by a second AI system only, so judge accuracy is supported by the dev set, judge–code agreement, the AI re-reads and the AI cross-check, not by human labels.
- The photo screening was done by AI (two systems) with the team lead deciding edge cases; no person reviewed every shipped photo. It removed 13 landmarks; 3 of the 4 removed for privacy were busy town-centre scenes, which are also among the easiest to recognise, so the removal took away 3 of the 70B's 9 correct place answers from photos (see H3).

Lessons Learnt and Recommendations

- For Apertus: abstention behaviour is language-dependent; calibration training should include Romansh and German, where the model guesses most. Visual recognition of Swiss heritage is the weakest link; grounding with retrieval (e.g. KGS/Wikidata lookup from GPS or image search) would help more than more text knowledge. The models already decline on uninformative photos; the risk is a photo that *looks* identifiable, so calibration data should include recognisable-looking but unknown objects.
- For evaluators: score abstention separately from accuracy and compare languages on identical images; check hedged answers ("not sure, but it seems to be a church") — they are abstentions, not errors.

Reproducibility

`make run` (Docker) on a clean checkout: builds the image, downloads the pinned dataset revision from Hugging Face, runs `verify` (schema, image hashes, coverage, re-grading reproduces every stored label, final-label rule, recomputed scores equal the expected scores), and, if `LLM_*` is set, evaluates that model on a stratified sample (`FULL=1` for all items). Hardware: any CPU; runtime ≈ 1 min for verification, under an hour for a full 70B run at 4 parallel requests. Seeds: selection 20261013, sampling 7, bootstrap 20261015, stability 20261016, audit 20261017. Dataset revision `26d86b35d1a6` (pinned in `config.json`); code: the `main` branch, tag `submission`.

License

Code: Apache-2.0. Dataset: CDLA-Permissive-2.0 (photos under their own CC licences, listed per item). Report: CC-BY-4.0.

References

SimpleQA (Wei et al., 2024); Wikidata; Swiss Inventory of Cultural Property of National and Regional Significance (KGS); BFS Swiss official commune register; Apertus technical report (Swiss AI Initiative, 2025).